

in-class-code

Packages

```
# general use  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.1.4      v readr      2.1.6  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2    4.0.2      v tibble     3.3.0  
v lubridate  1.9.4      v tidyr      1.3.1  
v purrr      1.2.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
# file organization  
library(here)
```

here() starts at C:/Users/hende/Documents/github/193-group-project

```
# data visualization  
library(naniar)
```

Warning: package 'naniar' was built under R version 4.5.3

```
library(patchwork)
```

Warning: package 'patchwork' was built under R version 4.5.3

Data

```
veg <- read_csv(here("data", "veg.csv"))
```

Rows: 27036 Columns: 17

-- Column specification -----

Delimiter: ","

chr (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...

dbl (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...

date (1): date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

Rows: 257 Columns: 4

-- Column specification -----

Delimiter: ","

chr (2): year_pool, transect_name

dbl (2): year, num_quad

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Class code

```
# create a new object from veg
gl_implicitNA <- veg |>
  # filter all occurrences where transect_name contains GL
  filter(str_detect(transect_name, "GL") &
```

```

    site == "ncos") |>
# group by year, pool, and species
group_by(year, transect_name, psoc) |>
# calculate mean percent cover
summarize(mean_pc = mean(percent_cover, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# create a new column called year_pool from year and transect name
unite("year_pool", year, transect_name, remove = TRUE) |>
# join with metadata by year_pool
left_join(metadata, by = "year_pool")

```

`summarise()` has grouped output by 'year', 'transect_name'. You can override using the `.groups` argument.

```

# creating new object from veg
gl_explicit0 <- veg |>
# filter all occurrences where transect_name contains GL
filter(str_detect(transect_name, "GL") &
    site == "ncos") |>
# complete all possible combinations of year, pool, psoc and distance
complete(year, transect_name, psoc, transect_distance,
    # fill values of percent cover with 0
    fill = list(percent_cover = 0)
) |>
# group by year, pool, and psoc
group_by(year, transect_name, psoc) |>
# calculate total percent cover across all quads
summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
# ungroup the data frame
ungroup() |>
# create a new column called year_pool from year and transect name
unite("year_pool", year, transect_name, remove = TRUE) |>
# join with metadata data frame
left_join(metadata, by = "year_pool") |>
# calculate mean percent cover from total percent cover/number of quadrats
mutate(mean_pc = sum_pc/num_quad)

```

`summarise()` has grouped output by 'year', 'transect_name'. You can override using the `.groups` argument.

```

# creating new object from veg
gl_native_explicit0 <- veg |>
# filter all occurrences where transect_name contains GL & native
filter(str_detect(transect_name, "GL") &
       site == "ncos" &
       cover_category == "NATIVE COVER") |>
# complete all possible combinations of year, pool, psoc and distance
complete(year, transect_name, psoc, transect_distance,
         # fill values of percent cover with 0
         fill = list(percent_cover = 0)
        ) |>
# group by year, pool, and psoc
group_by(year, transect_name, psoc) |>
# calculate total percent cover across all quads
summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
# ungroup the data frame
ungroup() |>
# create a new column called year_pool from year and transect name
unite("year_pool", year, transect_name, remove = TRUE) |>
# join with metadata data frame
left_join(metadata, by = "year_pool") |>
# calculate mean percent cover from total percent cover/number of quadrats
mutate(mean_pc = sum_pc/num_quad)

```

`summarise()` has grouped output by 'year', 'transect_name'. You can override using the `.groups` argument.

```

# creating new object from veg
gl_invasive_explicit0 <- veg |>
# filter all occurrences where transect_name contains GL & native
filter(str_detect(transect_name, "GL") &
       site == "ncos" &
       cover_category == "NON-NATIVE COVER") |>
# complete all possible combinations of year, pool, psoc and distance
complete(year, transect_name, psoc, transect_distance,
         # fill values of percent cover with 0
         fill = list(percent_cover = 0)
        ) |>
# group by year, pool, and psoc
group_by(year, transect_name, psoc) |>
# calculate total percent cover across all quads

```

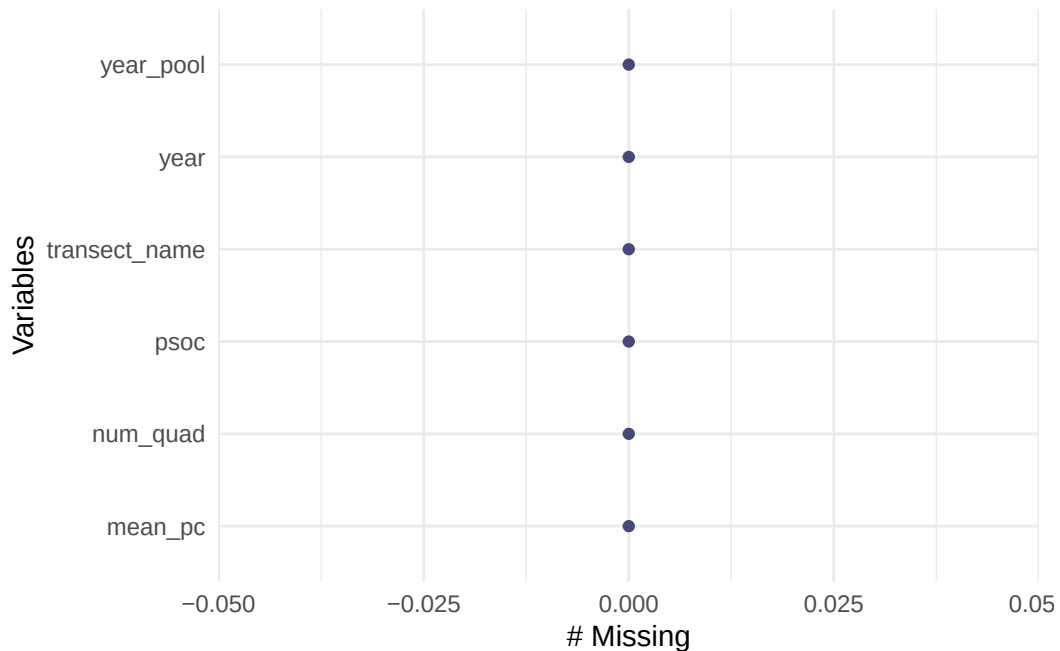
```

summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
# ungroup the data frame
ungroup() |>
# create a new column called year_pool from year and transect name
unite("year_pool", year, transect_name, remove = TRUE) |>
# join with metadata data frame
left_join(metadata, by = "year_pool") |>
# calculate mean percent cover from total percent cover/number of quadrats
mutate(mean_pc = sum_pc/num_quad)

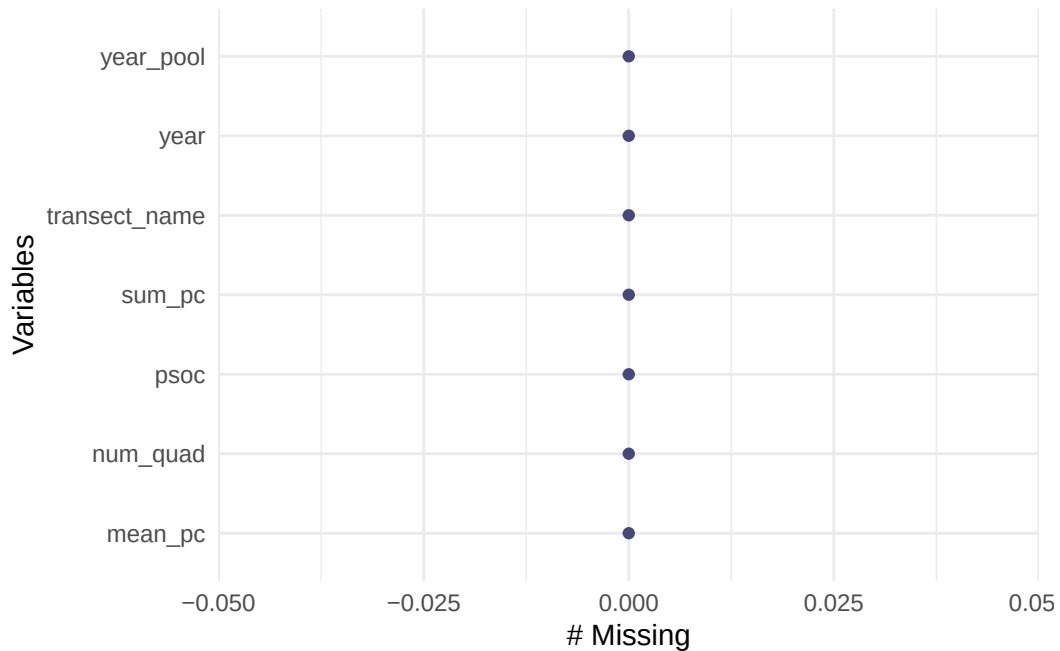
```

`summarise()` has grouped output by 'year', 'transect_name'. You can override using the `.groups` argument.

```
gg_miss_var(gl_implicitNA)
```



```
gg_miss_var(gl_explicit0)
```



No missing values in either data frame.

Grasslands vectors

```
# creating new vector from veg
native_spp <- veg |>
  # filter all occurrences where transect_name contains GL
  filter(str_detect(transect_name, "GL") &
    site == "ncos" &
    cover_category == "NATIVE COVER") |>
# pull to extract data frame column as a vector
pull(psoc)

# sample of ten rows
sample(native_spp, 10)
```

[1] "Stipa_pulchra"	"Stipa_pulchra"
[3] "Erigeron_canadensis"	"Stipa_pulchra"
[5] "Acmispon_maritimus"	"Heterotheca_grandiflora"
[7] "Deinandra_fasciculata"	"Symphyotrichum_subulatum"
[9] "Stipa_pulchra"	"Stipa_pulchra"

```
# creating new vector from veg
invasive_spp <- veg |>
  # filter all occurrences where transect_name contains GL and cover category is non-native
  filter(str_detect(transect_name, "GL") &
    site == "ncos" &
    cover_category == "NON-NATIVE COVER") |>
# pull to extract data frame column as a vector
pull(psoc)

# sample of ten rows
sample(invasive_spp, 10)
```

```
[1] "Bromus_hordeaceus"      "Festuca_perennis"
[3] "Brachypodium_distachyon" "Bromus_hordeaceus"
[5] "Lysimachia_arvensis"    "Bromus_hordeaceus"
[7] "Centaurium_tenuiflorum" "Trifolium_hirtum"
[9] "Helminthotheca_echioides" "Avena_fatua"
```

Species richness

```
# create new object from gl_explicit0
native_species_rich <- gl_explicit0 |>
# group by year and pool
group_by(year, transect_name) |>
# filter data frame with vector into get native species,
# also filter percent cover to identify species presence
filter(psoc %in% native_spp, mean_pc > 0) |>
# species richness calculation
summarize(spp_rich = n_distinct(psoc)) |>
# ungroup data frame
ungroup()
```

`summarise()` has grouped output by 'year'. You can override using the
`.groups` argument.

```
# sample 10 rows
slice_sample(native_species_rich, n = 10)
```

```
# A tibble: 10 x 3
  year transect_name spp_rich
  <dbl> <chr>         <int>
1  2024 GL-02             10
2  2023 GL-07              7
3  2021 GL-10              6
4  2022 GL-12              9
5  2024 GL-12              7
6  2021 GL-15              7
7  2021 GL-14             11
8  2024 GL-14             10
9  2021 GL-12             11
10 2021 GL-09             10
```

```
# create new object from gl_explicit0
invasive_species_rich <- gl_explicit0 |>
# group by year and pool
group_by(year, transect_name) |>
# filter data frame with vector into get native species,
# also filter percent cover to identify species presence
filter(psoc %in% invasive_spp, mean_pc > 0) |>
# species richness calculation
summarize(spp_rich = n_distinct(psoc)) |>
# ungroup data frame
ungroup()
```

`summarise()` has grouped output by 'year'. You can override using the
 `groups` argument.

```
# sample 10 rows
slice_sample(invasive_species_rich, n = 10)
```

```
# A tibble: 10 x 3
  year transect_name spp_rich
  <dbl> <chr>         <int>
1  2022 GL-02             21
2  2025 GL-15             10
3  2021 GL-12             18
4  2024 GL-07             16
5  2025 GL-07             10
6  2025 GL-09             16
```


7	2024	GL-11	22
8	2021	GL-11	9
9	2024	GL-15	14
10	2023	GL-11	13

Outliers???

Total Cover

```
# create new object from veg
native_pc <- veg |>
  # filter only native species
  filter(psoc %in% native_spp) |>
  # filter only grasslands at NCOS
  filter(str_detect(transect_name, "GL") & site == "ncos") |>
  # group by year and transect
  group_by(year, transect_name) |>
  # calculate implicit missingness sum
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # united column to match the one in metadata
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join metadata
  left_join(metadata, by = "year_pool") |>
  # dividing total % cover by the # of quadrats in the observed year
  mutate(total_pc = sum_pc/num_quad)
```

`summarise()` has grouped output by 'year'. You can override using the
`.groups` argument.

```
# sample of 10
slice_sample(native_pc, n = 10)
```

```
# A tibble: 10 x 6
  year_pool sum_pc year transect_name num_quad total_pc
  <chr>      <dbl> <dbl> <chr>          <dbl>      <dbl>
1 2025_GL-02    57  2025 GL-02          10        5.7
```

2	2024_GL-14	1088	2024	GL-14	11	98.9
3	2022_GL-10	671	2022	GL-10	11	61
4	2022_GL-11	370	2022	GL-11	11	33.6
5	2024_GL-11	655	2024	GL-11	11	59.5
6	2025_GL-10	296	2025	GL-10	11	26.9
7	2022_GL-07	637	2022	GL-07	11	57.9
8	2025_GL-15	173	2025	GL-15	11	15.7
9	2021_GL-02	346	2021	GL-02	11	31.5
10	2022_GL-02	500	2022	GL-02	11	45.5

```
# create new object from veg
invasive_pc <- veg |>
  # filter only invasive species
  filter(psoc %in% invasive_spp) |>
  # filter only grasslands at NCOS
  filter(str_detect(transect_name, "GL") & site == "ncos") |>
  # group by year and transect
  group_by(year, transect_name) |>
  # calculate implicit missingness sum
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # united column to match the one in metadata
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join metadata
  left_join(metadata, by = "year_pool") |>
  # dividing total % cover by the # of quadrats in the observed year
  mutate(total_pc = sum_pc/num_quad)
```

`summarise()` has grouped output by 'year'. You can override using the
`.groups` argument.

```
# sample of 10
slice_sample(invasive_pc, n = 10)
```

```
# A tibble: 10 x 6
  year_pool  sum_pc  year transect_name num_quad total_pc
  <chr>      <dbl> <dbl> <chr>          <dbl>    <dbl>
1 2025_GL-15    682  2025 GL-15             11      62
2 2025_GL-09   1022  2025 GL-09             11     92.9
3 2021_GL-12    395  2021 GL-12             11     35.9
```

4	2023_GL-12	982	2023	GL-12	11	89.3
5	2022_GL-12	364	2022	GL-12	11	33.1
6	2022_GL-02	288	2022	GL-02	11	26.2
7	2024_GL-07	1490	2024	GL-07	11	135.
8	2024_GL-10	1523	2024	GL-10	11	138.
9	2022_GL-10	419	2022	GL-10	11	38.1
10	2021_GL-10	201	2021	GL-10	11	18.3

Interest species wrangling

```
# native species
# species most significant: Stipa_pulchra and Distichlis_spicata
native_interest <- c(
  "Stipa_pulchra",
  "Distichlis_spicata",
  "Deinandra_fasciculata",
  "Erigeron_canadensis",
  "Symphyotrichum_subulatum"
)

# invasive / non-native species
# species most significant: Festuca_perennis and Polypogon_monspeliensis
non_native_interest <- c(
  "Festuca_perennis",
  "Polypogon_monspeliensis",
  "Bromus_hordeaceus",
  "Festuca_myuros",
  "Medicago_polymorpha"
)

# all selected species
all_interest <- c(
  "Stipa_pulchra",
  "Distichlis_spicata",
  "Deinandra_fasciculata",
  "Erigeron_canadensis",
  "Symphyotrichum_subulatum",
  "Festuca_perennis",
  "Polypogon_monspeliensis",
  "Bromus_hordeaceus",
```

```
"Festuca_myuros",
"Medicago_polymorpha"
)
```

```
gl_native_interest <- gl_native_explicit0 |>
  filter(psoc %in% native_interest) |>
  mutate(native = "native")

gl_invasive_interest <- gl_invasive_explicit0 |>
  filter(psoc %in% non_native_interest) |>
  mutate(native = "non_native")

gl_all_interest <- bind_rows(gl_native_interest, gl_invasive_interest)
```

```
# data frame for vis 1
gl_all_pc <- gl_all_interest |>
  group_by(psoc, year, native) |>
  summarize(sum_pc = sum(sum_pc),
            sum_quad = sum(num_quad)) |>
  ungroup() |>
  mutate(mean_pc = sum_pc/sum_quad)
```

`summarise()` has grouped output by 'psoc', 'year'. You can override using the `.groups` argument.

```
# tbd
```

```
# data frames for vis 3
gl_native_tc <- gl_native_interest |>
  group_by(psoc, year_pool, native) |>
  summarize(sum_pc = sum(sum_pc)) |>
  ungroup()
```

`summarise()` has grouped output by 'psoc', 'year_pool'. You can override using the `.groups` argument.

```
gl_invasive_tc <- gl_invasive_interest |>
  group_by(psoc, year_pool, native) |>
  summarize(sum_pc = sum(sum_pc)) |>
  ungroup()
```

``summarise()`` has grouped output by 'psoc', 'year_pool'. You can override using the ``.groups`` argument.

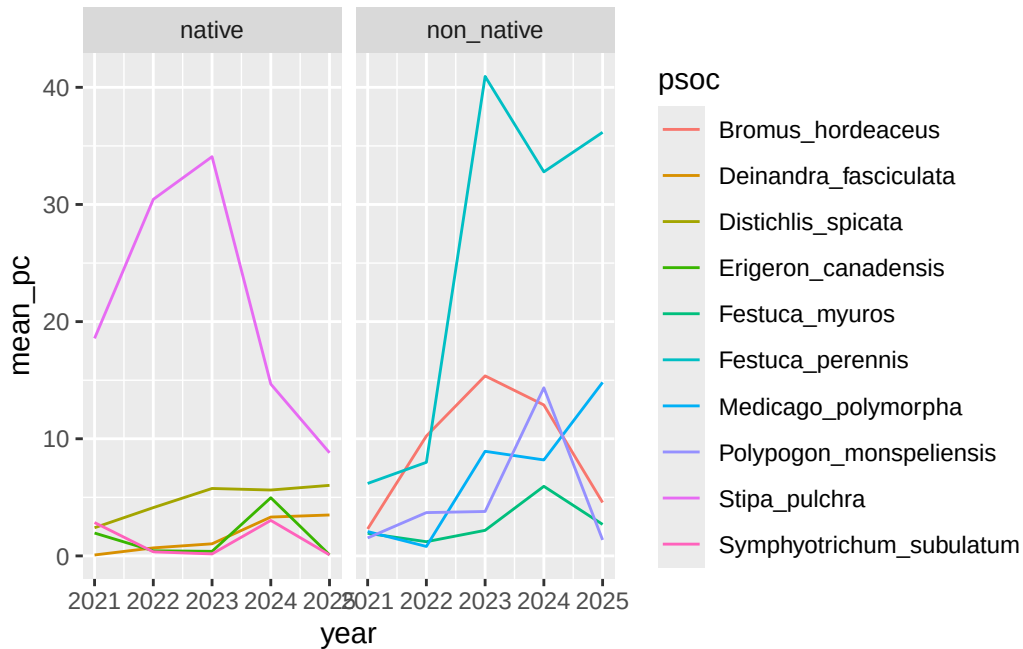
```
gl_stacked_native <- gl_all_interest |>
  group_by(native, year) |>
  summarize(mean_pc_sum = sum(mean_pc)) |>
  ungroup() |>
  mutate(native)
```

``summarise()`` has grouped output by 'native'. You can override using the ``.groups`` argument.

Visualizations

```
vis1 <- ggplot(data = gl_all_pc,
  aes(x = year,
      y = mean_pc,
      color = psoc)) +
  geom_line() +
  facet_wrap(~ native)

vis1
```



```
native_tc_plot <- ggplot(data = gl_native_tc,
  aes(x = psoc,
    y = sum_pc,
    color = psoc)) +
  geom_jitter(height = 0,
    width = 0.2) +
  stat_summary(geom = "pointrange",
    # argument to calculate standard error
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    size = 1,
    linewidth = 1,
    color = "red") +
  labs(title = "Native Total Cover") +
  theme(legend.position = "none")

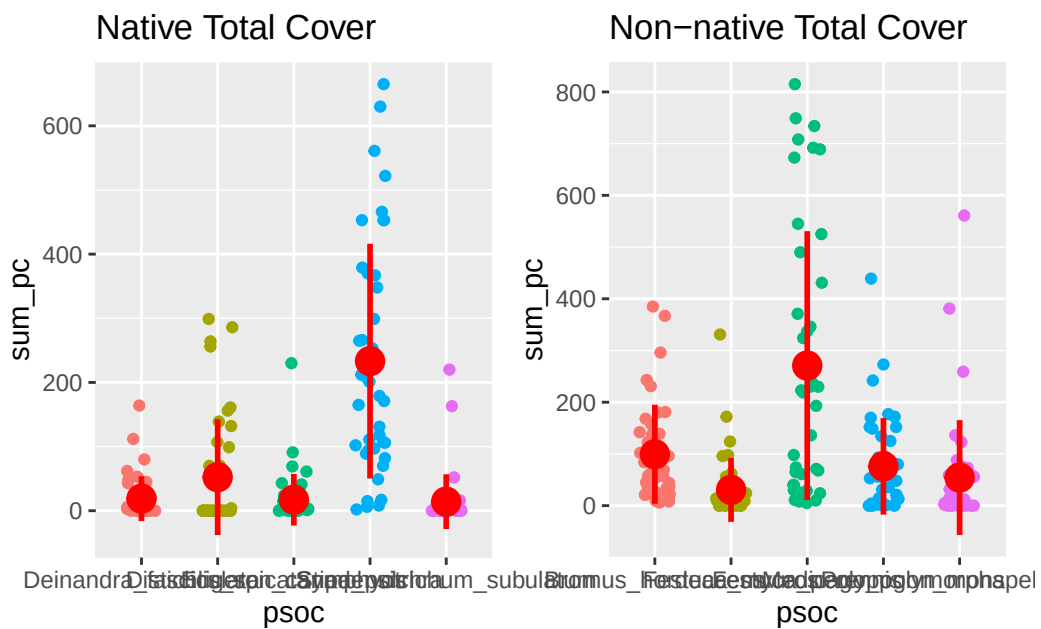
invasive_tc_plot <- ggplot(data = gl_invasive_tc,
  aes(x = psoc,
    y = sum_pc,
    color = psoc)) +
  geom_jitter(height = 0,
    width = 0.2) +
  stat_summary(geom = "pointrange",
```

```

# argument to calculate standard error
fun.data = mean_sdl,
fun.args = list(mult = 1),
size = 1,
linewidth = 1,
color = "red") +
labs(title = "Non-native Total Cover") +
theme(legend.position = "none")

vis3 <- native_tc_plot + invasive_tc_plot
vis3

```



```

vis4 <- ggplot(data = gl_stacked_native,
  aes(x = year,
    y = mean_pc_sum,
    fill = native)) +
  geom_area()

vis4

```

